Every job starts the same way: a drawing in an email.

DR

Dana Reyes <dana@northwindrobotics.com>
to quotes@ · 8:02 AM

RFQ: Sensor housing rev B, 25 pcs

Hi team, please quote 25 pcs of the attached housing in 6061-T6.
We need parts in 3 weeks. Can you get us a price by Friday?

Thanks, Dana

PDF housing_rev-B.pdf

STEP housing_rev-B.step

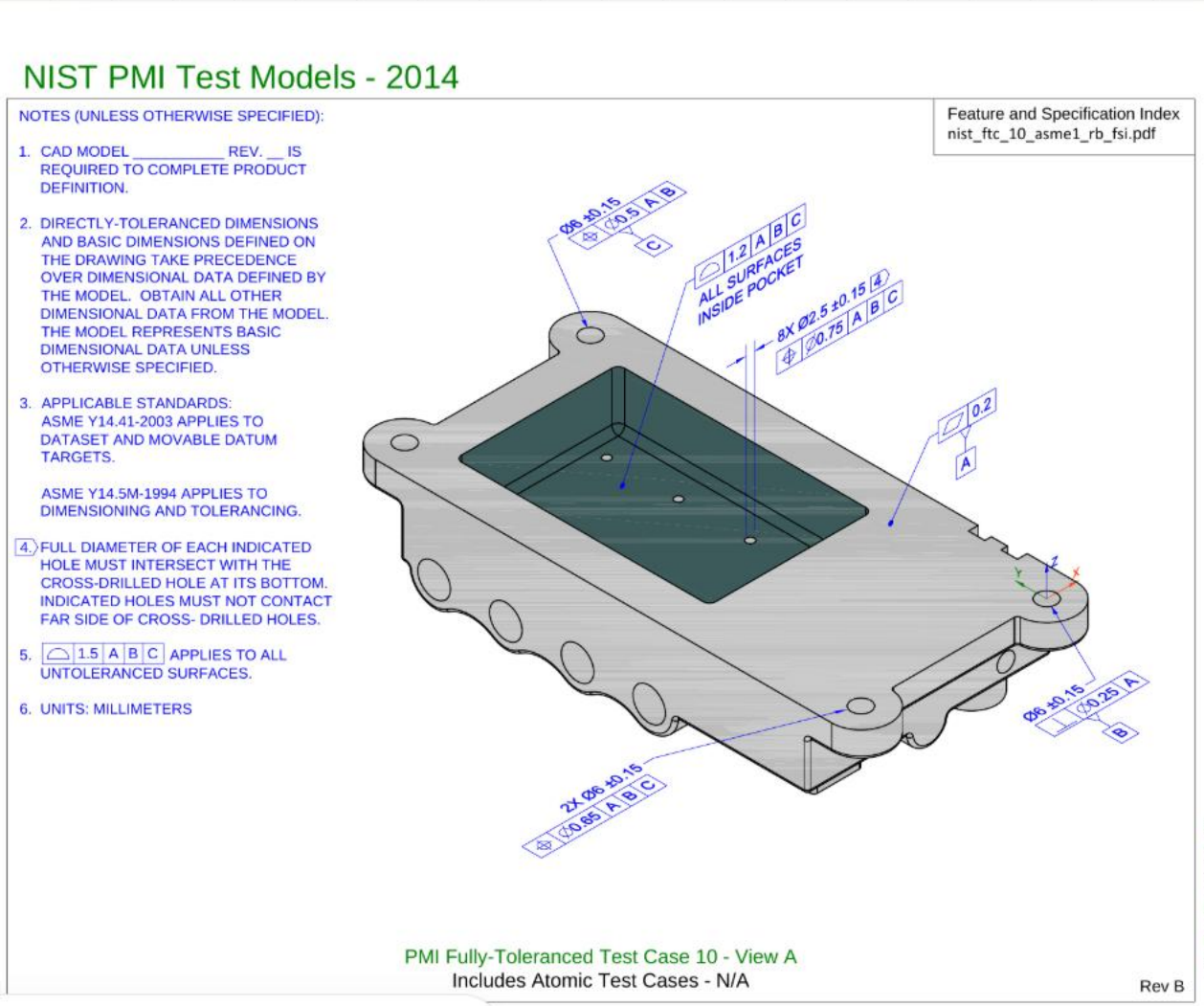
Lakeside Medical Thu
RFQ: manifold block, 10 pcs

Arcwell Drones Thu
RFQ: motor bracket rev C, 200 pcs

Halvorsen Marine Fri
RFQ: pump cover, 4 pcs

“A build-to-print vendor gets a drawing, quotes it, and makes the part.”

a16z



housing_rev-B.pdf · Qty 25 · 6061-T6

Every hole toleranced to ±0.15 mm.

From email to finished part.

RFQ email
drawing + 3D model

Quote
price & lead time

CAM programming
toolpaths & G-code
4–8 h
expert only

Setup
fixtures & tools

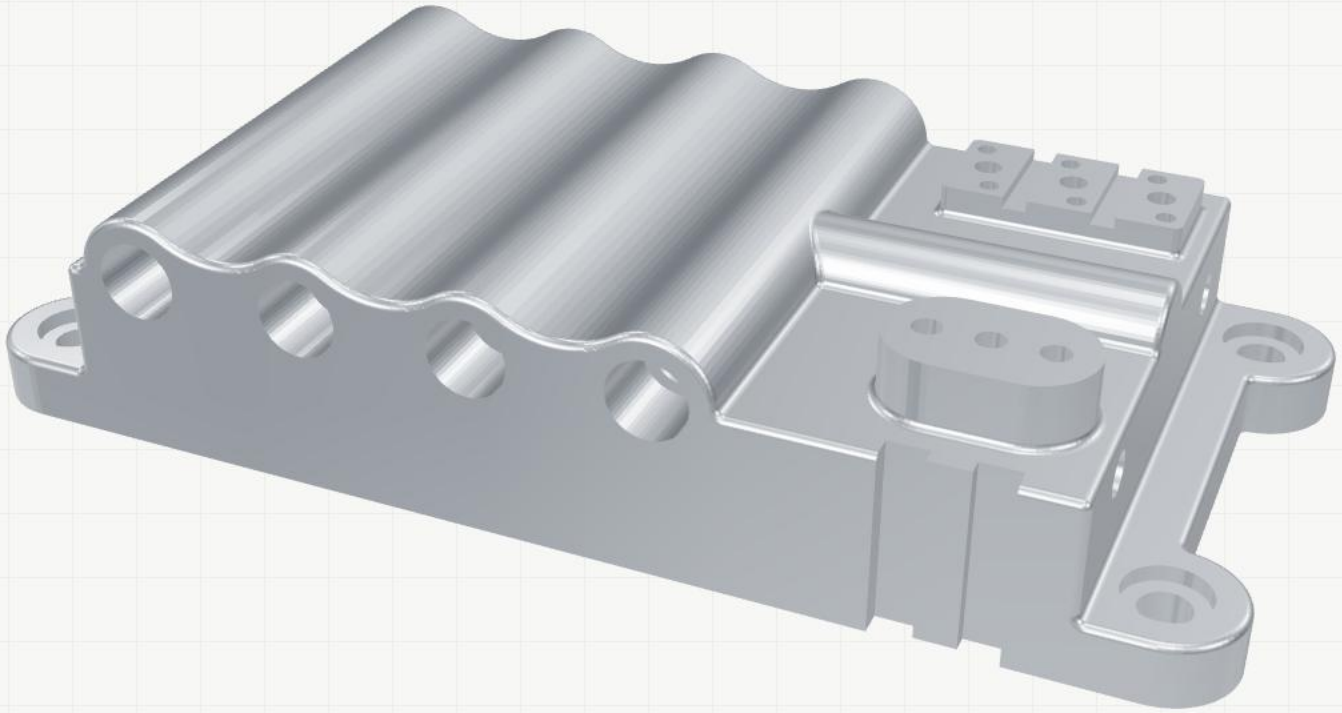
Machining
the actual cutting

Inspection
check tolerances

Ship
to the customer

Every new part passes through one expert before a machine can start.

Before a single chip is cut, someone has to program it.



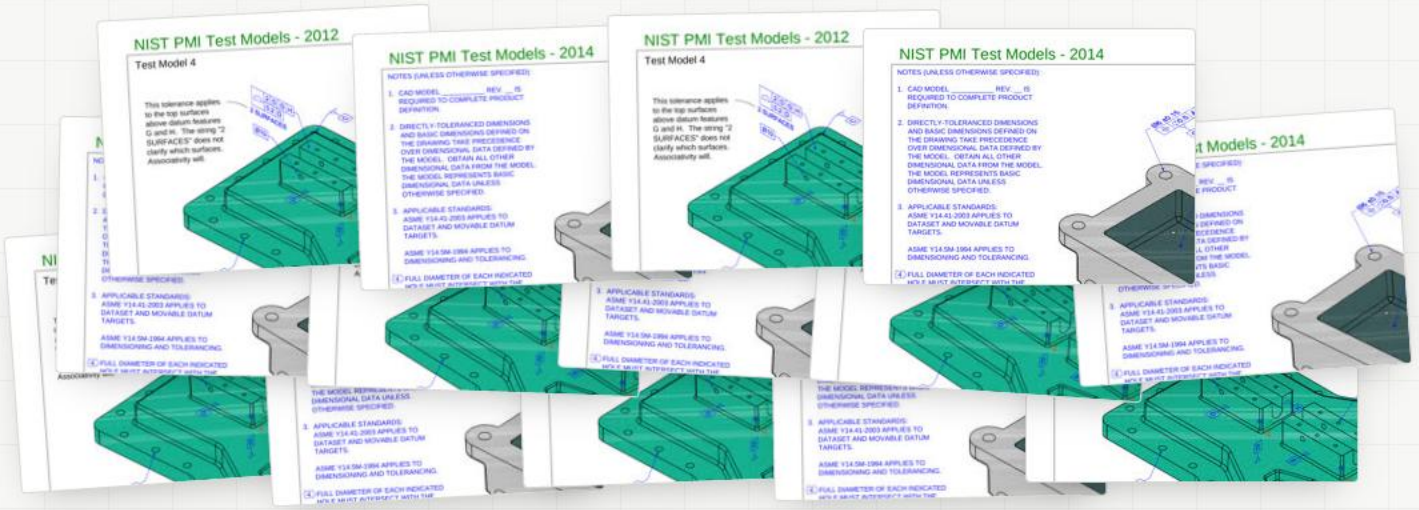
4:15

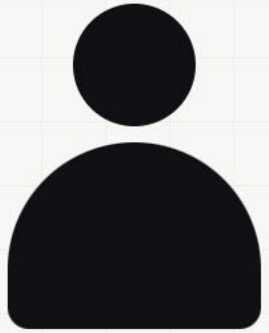
hours of CAM programming

✓ Import model & drawing	0:05
✓ Stock, fixturing, work offsets	0:25
✓ Pick tools & holders	0:50
✓ Program 20+ toolpaths	2:45
✓ Simulate, fix collisions	3:40
✓ Post & prove out G-code	4:15

4–8 hours for one new part · Modern Machine Shop

Every new part waits for the one person who can program it.





Lead machinist

Average age: **mid-50s**

a16z · Investing in Hadrian

<10%

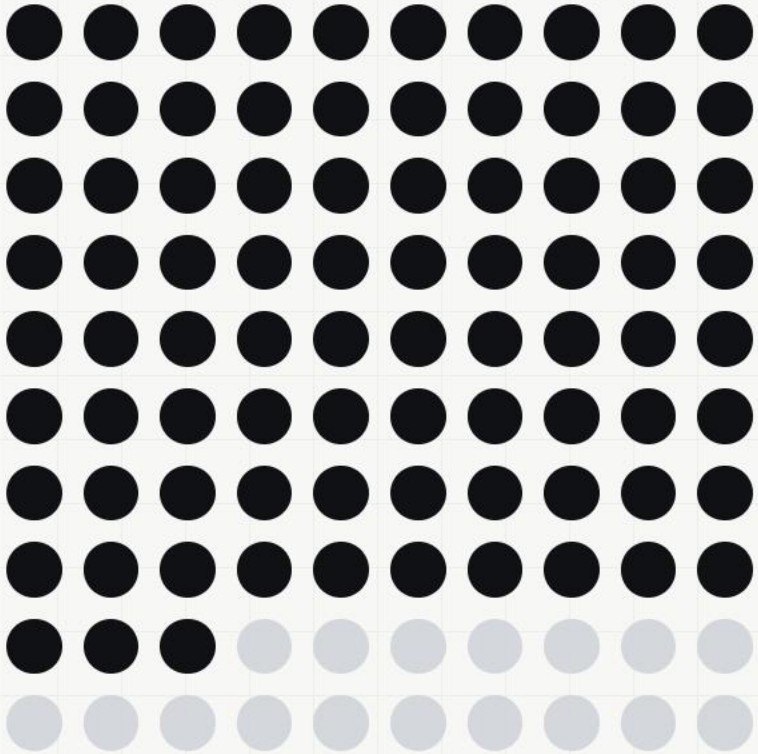
of a part's time in the factory is spent being made.
The rest is waiting.



16,876

US machine shops.
83% have fewer than 20 people.

a16z · The Case for the American Manufacturing Asset Class



“Most are operating at a fraction of their potential because of dated software, engineering, workflows, and equipment.”

a16z · The Case for the American Manufacturing Asset Class

~~4–8 h~~ → **minutes**

and **shorter** machine time on every part

Instant Machinist

Drawing in. Proven CNC program out.